

Bernstein's polynomial inequality on infinite-dimensional complex L_p spaces

Literature-implied partial answer for arXiv:2003.11002

21 August 2026

Status. **Literature-implied answer, complete for the infinite-dimensional complex $L_p(\mu)$ subfamily.** This is an identification of an older theorem, not a new proof and not a characterization of arbitrary complex Banach spaces.

Source problem

Chatzakou and Sarantopoulos ask in Problem 17 of arXiv:2003.11002 (official PDF page 17) for a characterization of the complex Banach spaces X such that every continuous m -homogeneous polynomial $P = \hat{L}$ satisfies

$$\|DP\| \leq m\|P\|, \quad \text{equivalently} \quad \|L\| = \|P\|,$$

for every degree m [1]. In the usual notation this says $c(m, X) = 1$ for all $m \geq 1$.

Supporting theorem and identification

Dimant, Galicer, and Rodriguez call $c(2, X) = 1$ the symmetric operator norm property. Their Theorem 3.1 states that an infinite-dimensional complex Banach space with this property must satisfy

$$p(X) = q(X) = 2,$$

where $p(X)$ and $q(X)$ are its optimal type and cotype exponents [2, Theorem 3.1].

Literature-implied subfamily classification. *Let $L_p(\mu)$ be infinite dimensional and complex, with $1 \leq p \leq \infty$. Then the following are equivalent:*

1. $c(m, L_p(\mu)) = 1$ for every $m \geq 1$;
2. Bernstein's inequality $\|DP\| \leq m\|P\|$ holds for every continuous homogeneous polynomial and every degree;
3. $p = 2$.

Identification. The standard optimal exponents of an infinite-dimensional L_p space are

$$p(L_p) = \min\{p, 2\}, \quad q(L_p) = \max\{p, 2\},$$

with the usual endpoint interpretation. The all-degree property implies $c(2, L_p) = 1$, so the supporting theorem forces both exponents to equal two and hence $p = 2$. Conversely, $L_2(\mu)$ is a complex Hilbert space and Banach's polarization theorem gives $c(m, L_2) = 1$ in every degree. The source records the derivative/polarization equivalence, giving $(1) \Leftrightarrow (2)$.

Provenance and scope

The supporting paper predates the source problem and does not identify itself as an answer to Problem 17. The relation is therefore an agent-identified implication. It settles only the infinite-dimensional complex L_p family. It does not classify finite-dimensional spaces, arbitrary complex Banach spaces with $c(2, X) = 1$, or spaces $H_1 \oplus_\infty H_2$.

The official source page and supporting Theorem 3.1 (including its proof) were independently inspected rather than inferred from the interrupted temporary PDF. Exact-id, theorem-name, polarization-constant, and type/cotype searches found no full arbitrary-space classification through 21 August 2026. A separate attempt record documents the unsuccessful full upgrade.

References

- [1] M. Chatzakou and Y. Sarantopoulos, *Bernstein and Markov-type inequalities for polynomials on $L_p(\mu)$ spaces*, arXiv:2003.11002 (2020), Problem 17.
- [2] V. Dimant, D. Galicer, and J. T. Rodriguez, *The polarization constant of finite dimensional complex spaces is one*, arXiv:1908.08107 (2019), Theorem 3.1.